

Question 1: **8.0 (option b)**

Question 2: **65 66 67 68 69 (option b)**

Question 3: **0.6666666666666666 (option d)**

Question 4: **Compiler error (option e)**

Question 5: **30 (option c)**

Question 6:

```
... public static String encrypt(String str){
...     String result="";
...     int len = str.length();
...     for (int i = 0; i < len; i++){
...         char ch = str.charAt(i);
...         char shifted_char = (char) ('a' + (ch - 'a' + 2) % 26);
...         result += shifted_char;
...     }
...     return result;
... }
```

Question 7:

```
public static void reverse(String str){
→ // prints the string in reverse manner.
→ int len = str.length();
→ // as we need to the print the string in reverse manner traversing the string from last index to first
→ for (int i = len-1; i >= 0; i--){
→     System.out.print(str.charAt(i));
→ }
→ System.out.println();
}
```

Question 8:

```
1  import java.util.Scanner;
2
3  public class TriangleNumbers {
4      ... public static void main(String[] args) {
5          ... Scanner sc = new Scanner(System.in);
6          ... System.out.println(x: "Enter an integer");
7          ... int x = sc.nextInt();
8          ... int n = 1;
9          ... int line = 1;
10         ... while(n<=x){
11             ... for(int i=0; i<genNum(line) && n<=x; i++){
12                 ... System.out.print(genNum(n) + ".");
13                 ... n++;
14             ... }
15             ... System.out.println();
16             ... line++;
17         ... }
18         ... sc.close();
19     ... }
20
21     ... public static int genNum(int n){
22         ... int triangleNumber = 0;
23         ... for (int i=1; i<=n; i++){
24             ... triangleNumber+=i;
25         ... }
26         ... return triangleNumber;
27     ... }
28 }
```

Question 9:

```
import java.util.Scanner;

public class HollowSandglass {
    private static Scanner sc;
    public static void main(String[] args) {
        int i, j, k;
        sc = new Scanner(System.in);
        System.out.print("Enter the hourglass Pattern Rows = ");
        int rows = sc.nextInt();

        System.out.println("Printing hourglass Pattern");

        // Adjust the rows for even and odd inputs
        // For symmetry and to match the number of rows
        if(rows % 2 == 1)
            rows = (rows / 2) + 1;
        else
            rows = rows / 2;

        // Print the upper half
        for (i = 1; i <= rows; i++) {
            for (j = 1; j < i; j++) {
                System.out.print(" ");
            }
            for (k = i; k <= rows; k++) {
                if (i == 1 || k == i || k == rows) {
                    System.out.print("* ");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

```

// Print the bottom half
for (i = rows - 1; i >= 1; i--) {
    for (j = 1; j < i; j++) {
        System.out.print(" ");
    }
    for (k = i; k <= rows; k++) {
        if (i == 1 || k == i || k == rows) {
            System.out.print("* ");
        } else {
            System.out.print(" ");
        }
    }
    System.out.println();
}
}
}

```

Question 10:

```

import java.util.Scanner;

public class Q10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter an integer:");
        int n = sc.nextInt();
        // finding sum of all multiples of 3 or 5 - Make sure you don't add multiples of 3 and 5 twice,
        // for ex 15 should be added only once.
        int sum = 0;
        for (int i = 1; i <= n; i++) {
            if ((i % 3 == 0) || (i % 5 == 0)) {
                sum += i;
            }
        }
        System.out.println("The sum of all multiples of 3 or 5 under "+n+" is "+sum);
        sc.close();
    }
}

```